Milestone 2 – Segmentation Performance Report

Artery Segmentation and Stenosis Detection Using Advanced Deep Learning Techniques

1. Objective

This report summarises how the segmentation models were trained and evaluated in Milestone 2, compares the baseline U-Net with the Attention U-Net, and states which model was selected for the application and why.

2. Dataset and Tasks

The ARCADE dataset (2D X-ray coronary angiography) was used. It provides two segmentation tasks, each with its own train, validation and test split:

- **Syntax task:** segmentation of the coronary artery segments.
- **Stenosis task:** segmentation of narrowed (stenotic) regions. Stenosis split sizes: 1000 training, 200 validation and 300 test images.

3. Models and Training Setup

Item	Setting
Baseline model	U-Net (base 32 filters)
Advanced model	Attention U-Net (attention gates on the skip connections)
Input size	256 × 256
Batch size / learning rate	8 / 1e-3
Epochs	30
Hardware	Google Colab, T4 GPU
Prediction threshold	0.5
Best checkpoint	Chosen by highest validation Dice

Both tasks were trained separately for each architecture. Inference inside the application runs on CPU.

4. Evaluation Metrics

- **Dice coefficient:** overlap between predicted and true mask, $2\cdot|A\cap B| / (|A|+|B|)$. The main metric.
- **IoU (Jaccard):** intersection over union, a stricter overlap measure.
- **Precision:** of the pixels predicted as target, how many are correct.
- **Recall:** of the true target pixels, how many were found.
- **Specificity:** how well background pixels are correctly left out.
- **Pixel accuracy:** share of all pixels classified correctly. It is high even for poor models because background dominates, so Dice and IoU matter more.

5. Results

5.1 Baseline U-Net (model used in the application)

Metric	Syntax	Stenosis
Best validation Dice	0.8029	0.506
Test Dice	0.7597	0.4888
Test IoU	0.6126	0.3235
Precision	0.7788	0.5159
Recall	0.7416	0.4644
Specificity	0.9922	–
Pixel accuracy	0.9832	–
Mean Dice per image	–	0.4453

The stenosis row is the retrained model (256 px, 30 epochs). An earlier first run reached a best validation Dice of 0.4509 and a test Dice of 0.4378, so the retraining improved it.

5.2 Attention U-Net

Metric	Syntax	Stenosis
Best validation Dice	0.7966	0.493
Test Dice	0.7509	0.478
Test IoU	0.6012	0.3142
Precision	0.7792	0.4805
Recall	0.7246	0.4759
Specificity	0.9924	0.9950
Pixel accuracy	0.9828	0.9900
Mean Dice per image	0.7437	0.4433

The syntax numbers come from a clean re-run of the notebook (restart, then run all). Reloading the saved checkpoint reproduced the training validation Dice of 0.7966, so the training and evaluation results agree.

5.3 Side-by-Side Comparison (test Dice)

Task	U-Net	Attention U-Net
Syntax	0.7597	0.7509
Stenosis	0.4888	0.478

6. Discussion

- **Syntax (artery segmentation):** both models reach a test Dice of about 0.75–0.76, which is a solid result. Arteries are long, continuous structures that cover a reasonable share of the image, so they are easier to learn.
- **Stenosis:** both models score around 0.48–0.49 Dice. Stenotic regions are very small, the classes are heavily imbalanced (specificity and pixel accuracy are high mainly because of the background), and annotations are subtle, so overlap scores stay lower.
- **Architecture comparison:** the attention gates gave no measurable gain on either task (syntax 0.7509 vs 0.7597, stenosis 0.478 vs 0.4888). The gaps are about 0.01 or less, so the two models perform almost the same on this dataset and training budget.

7. Selected Model

The U-Net was selected as the segmentation model for the application.

- Slightly higher test Dice on both tasks (syntax 0.7597 vs 0.7509, stenosis 0.4888 vs 0.478), although the margin is small.
- Simpler architecture with the same level of accuracy, so the extra attention gates bring no benefit here.
- Lightweight, so it runs on CPU inside the app (model version `unet-artery-v1`).

The Attention U-Net remains in the project as the advanced architecture that was implemented, trained and compared.

8. Segmentation Pipeline in the Application

After the model predicts a mask, the backend applies these steps:

1. **Post-processing:** removes small noise and edge artifacts (border guard and bar filter).
2. **Vessel structure extraction:** finds the connected vessel segments.
3. **Centerline extraction:** skeletonizes the vessels into a thin line along each vessel.
4. **Diameter estimation:** a distance transform along the centerline gives mean, maximum and minimum vessel diameter in pixels.

Results (mask, overlay, centerline image and vessel numbers) are saved to the database and can be reopened from the dashboard.

9. Limitations and Future Work

- Stenosis Dice (~0.49) can be improved with higher resolution (512 px), stronger augmentation, starting from the syntax weights, and threshold tuning on the validation set.
- Compare U-Net and Attention U-Net over several random seeds to check whether the small gap between them is stable.
- Diameters are in pixels. Converting to millimetres needs calibration data.
- Milestone 3 will build stenosis detection, severity grading and explainable AI on top of these masks.

10. Conclusion

The segmentation stage of Milestone 2 is complete. A U-Net and an Attention U-Net were trained and evaluated on both ARCADE tasks, and the U-Net was selected as the best-performing model, although the Attention U-Net performed almost the same. It reaches a test Dice of 0.7597 for artery segmentation and 0.4888 for stenosis, and it is integrated into the application together with post-processing, centerline extraction and diameter estimation.